

Topic 4, Part 1

Completely Randomized Design with Two Arms

PUBH 526

Design and Analysis of Randomized Trials in Public Health

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Completely Randomized Design (CRD)

- “Treatments” or “arms” or “groups” or “levels of a factor”
- Treatments happen in parallel (at the same time)
- One of the treatments may be a control or standard of care
- Participants (“experimental units”) are individually randomized to one of the treatments
 - What are the alternatives?



Side note: experimental designs differ by the **restrictions they put on randomization**

Randomization

- Randomly assigning experimental units to treatments:
 - Each experimental unit has the same allocation probabilities
 - All treatment allocations are equally likely
- Do not need to have equal numbers of experimental units in each treatment (the arms do not need to be **balanced**)
- Denoting response data:

Treatment 1 $y_{11}, y_{12}, \dots, y_{1n_1}$

Treatment 2 $y_{21}, y_{22}, \dots, y_{2n_2}$

Example

- Using the dataset “hb_2020_01_28” and the program “hb analysis v3.sas” on Brightspace
- Following Etheredge et al., we have data on hemoglobin at delivery for 200 women – half receiving iron supplements and half receiving a placebo
- “Random” comes up twice:
 - Selection of participants
 - Allocation of participants to treatments

Example

- Does taking iron during pregnancy affect hemoglobin at delivery?
 - (All participants were iron replete at randomization)
- Explore and check your data – get to know it
- Summary stats, graphs

If Someone is Accused of a Crime...	Hypothesis Testing
Did they commit the crime?	Does hemoglobin at delivery depend on whether a person takes iron supplements during pregnancy?
Starting assumption: nothing happened: they are innocent	Null hypothesis: nothing is happening: iron supplementation during pregnancy does not affect hemoglobin at delivery
Alternatively, they are guilty	Alternative hypothesis: iron does affect hemoglobin at delivery
Gather evidence	Gather data
How possible is it that they are innocent and all the evidence happened only by chance?	P-value! The probability that iron does not affect hemoglobin at delivery but you still see data as extreme as yours or even more extreme
Judgement: • “Guilty”: the evidence is very unlikely to happen only by chance • “Not Guilty”: the evidence is not strong enough – maybe it’s chance	Decision: • “Reject null hypothesis”: $p < \alpha$ (usually $\alpha = 0.05$, so $p < 0.05$) • “Do not reject null hypothesis”: $p \geq \alpha$ (usually $\alpha = 0.05$, so $p \geq 0.05$)
Does “Not Guilty” mean innocence?	Can we “accept” the null hypothesis?

Hypothesis Testing

1. State the scientific question to be answered

Does taking iron during pregnancy affect hemoglobin at delivery?

2. Define the parameters of interest

μ_1 = mean hemoglobin (in g/dL) at delivery among women (of our population) who take iron during pregnancy

μ_2 = mean hemoglobin (in g/dL) at delivery among women (of our population) who take a placebo/do not take iron during pregnancy

(parameter vs. estimate)

Hypothesis Testing

3. State the null hypothesis H_0 both mathematically (in terms of parameters) and in words

- Usually an equality

$$H_0: \mu_1 = \mu_2$$

Women in this population who take iron during pregnancy have the same mean hemoglobin at delivery as women in this population who do not take iron during pregnancy

4. Decide whether the test is directional or not, and state the alternative hypothesis (H_A or H_1) both mathematically and in words

$$H_A: \mu_1 \neq \mu_2 \text{ (not directional)}$$

In this population, the mean hemoglobin at delivery is different among women who take iron during pregnancy compared with women who do not take iron during pregnancy

Hypothesis Testing

5. Name the type of statistical test to be used and the distribution the test statistic follows under H_0

- Determine appropriate test statistic based on null hypothesis
- Include the degrees of freedom (or other parameters) for that distribution if appropriate

t test

t statistic

$t \mid H_0 \sim t_{n-2}$

$n=200$, so $df=198$

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Hypothesis Testing

6. State the significance level α to be used and the corresponding critical value of the test statistic. Define the rejection region.

- The critical value and rejection region depend on α , the df, the distribution from #5, and whether H_A is directional

$$\alpha = 0.05$$

Critical values: $t_{\alpha/2, df=n-2}$ and $t_{1-(\alpha/2), df=n-2}$

Critical values: $t_{0.025, df=198}$ and $t_{0.975, df=198}$

Critical values: -1.972 and 1.972

Reject H_0 if $t < -1.972$ or $t > 1.972$

← Draw a picture!*

* What exactly is this distribution?

Hypothesis Testing

7. Calculate the test statistic from the data
8. Compare the test statistic to the rejection region, or give the p-value and compare it to α .
 - P-value $<$ or $>$ α ?

$$t = 3.17$$

$$3.17 > 1.972$$

$$p = 0.0018 < 0.05$$

Hypothesis Testing

9. Make a decision about the null hypothesis:
 - a) If the test statistic is in the rejection region, we "reject H_0 "
 - b) If not, we "do not reject H_0 "

We DO NOT use the word "accept" anywhere in this step.
If using p-value, reject H_0 when $p < \alpha$.

We reject H_0

T-test Vs. ANOVA?

$$t_{df}^2 = F_{1,df}$$

Our t-test is also an ANOVA!

(so don't bother doing both – they are the same)

Hypothesis Testing

10. Form a scientific conclusion based on your decision.

If 9a), then start with “This study provides evidence...”

If 9b), then start with “This study does not provide evidence...”

followed by "at the ____ significance level that"

followed by the verbal statement of H_A .

This study provides evidence at the 0.05 significance level that in this population, the mean hemoglobin at delivery is different among women who take iron during pregnancy compared with women who do not take iron during pregnancy

More on the Confidence Interval

- δ = true difference in group means
- 100(1- α)% CI:

$$\hat{\delta} \pm t_{df, \alpha/2} SE(\hat{\delta})$$

- If underlying assumptions are true, δ will fall within 100(1- α)% of possible intervals
- You are 100(1- α)% confident that your single CI is one of those intervals that contain δ

More on the Confidence Interval

- Relation between CI and statistical test?

- If:

δ = true difference in group means

$$H_0: \mu_1 = \mu_2$$

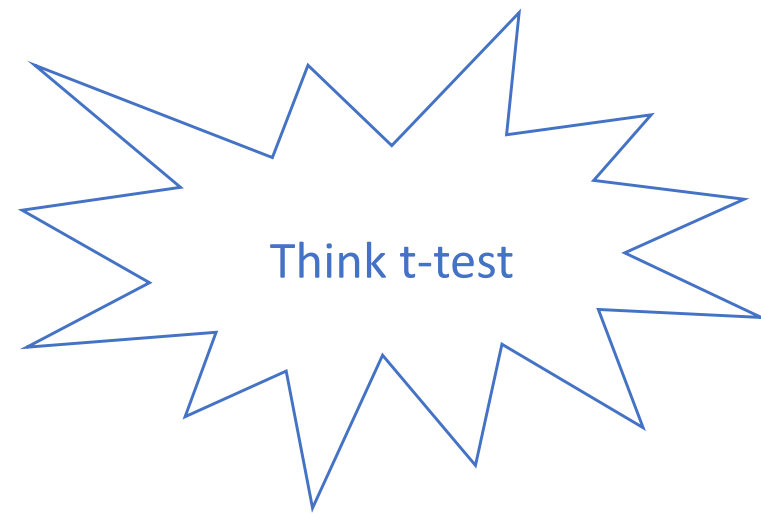
95% CI for the difference
in means:
(0.3274, 1.4086) and we
rejected H_0

- Then:

$$\delta = \text{true difference in group means} = \mu_1 - \mu_2 = 0$$

- If the 95% CI for δ contains 0 and you do a statistical test of H_0 at a 0.05 significance level, you will NOT reject
- If the 95% CI for δ does NOT contain 0 and you do a statistical test of H_0 at a 0.05 significance level, you will reject

Statistical Model



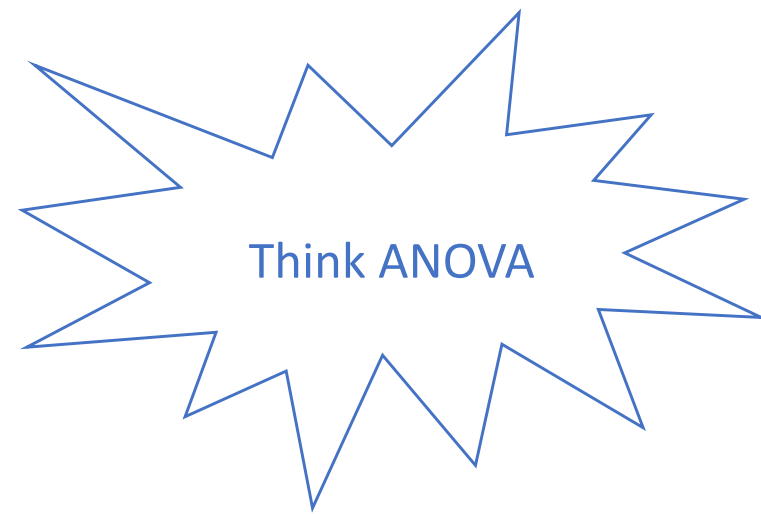
- Option 1

$$y_{ij} = \mu_i + \varepsilon_{ij} \begin{cases} i = 1, 2 \\ j = 1, 2, \dots, n_i \end{cases}$$

$$\varepsilon_{ij} \sim iid N(0, \sigma^2)$$

- μ_i = mean of group i
- $H_0: \mu_1 = \mu_2$
- Where in the model or analysis does the randomized design get factored in?

Statistical Model



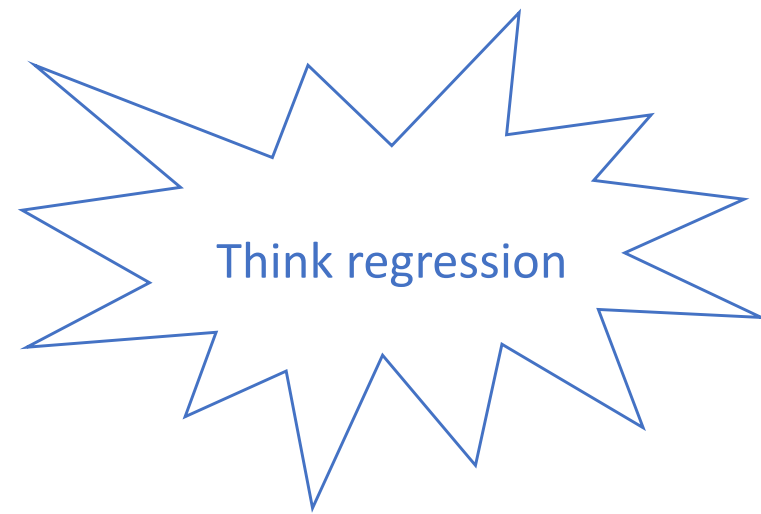
- Option 2

$$y_{ij} = \mu + \tau_i + \varepsilon_{ij} \begin{cases} i = 1, 2 \\ j = 1, 2, \dots, n_i \end{cases}$$

$$\varepsilon_{ij} \sim iid N(0, \sigma^2)$$

- μ = overall mean
- $\mu_i = \mu + \tau_i$
- τ_i = effect of treatment i
 - Note if there are two treatment arms, $\tau_1 = -\tau_2$
- $H_0: \tau_1 = \tau_2 = 0$

Statistical Model



- Option 3

$$y_j = \beta_0 + \beta_1 X + \varepsilon_j$$

$$\varepsilon_j \sim iid N(0, \sigma^2)$$

- $j = 1, 2, \dots, \Sigma n_i$ (full sample, across all groups)
- $X=1$ if observation is in the intervention group,
0 otherwise
 - Here X is an *indicator function*: it indicates whether an observation is in the intervention group
- $H_0: \beta_1 = 0$
- $\widehat{\beta}_0$ = mean in the control group